

# Retail Store Sales & Customer Behavior Analysis

**TEYZIX CORE Internship Program**

**Intern ID:** TC-INT-20260603-498

**Task ID:** DA-2

**Domain:** Data Analytics

## 1. Project Overview

Retail businesses generate thousands of transactions every day. However, raw sales data alone cannot support business decision-making unless it is cleaned, analyzed, and visualized. This project was developed to perform a complete end-to-end data analytics workflow using a self-created retail sales dataset. The project includes dataset creation, data cleaning, exploratory data analysis (EDA), business insight generation, and dashboard development using Google Colab (Python) and Microsoft Power BI.

## 2. Objectives

The objectives of this project are:

- Create a realistic retail sales dataset with more than 1,000 records.
- Clean and preprocess the raw dataset.
- Perform Exploratory Data Analysis (EDA).
- Identify customer purchasing behavior and sales trends.
- Develop an interactive Power BI dashboard.
- Generate business insights and recommendations.

## 3. Dataset Description

A self-created dataset containing **1,200 retail transactions** was developed.

The dataset contains the following fields:

- Transaction ID
- Customer ID
- Purchase Date
- Customer Name
- Gender
- Age
- City
- Product Category
- Product Name

- Quantity
- Unit Price
- Discount
- Total Amount
- Payment Method

The dataset represents sales transactions from different cities of Pakistan and includes multiple product categories such as Electronics, Accessories, Home Appliances, Gaming, and Office products.

## 4. Tools & Technologies

The following tools were used during this project:

- Google Colab
- Python
- Pandas
- NumPy
- Matplotlib
- Microsoft Power BI

## 5. Data Cleaning Process

The raw dataset contained several quality issues that were cleaned before analysis.

The following cleaning steps were performed:

### 5.1 Missing Values

Missing values were identified in:

- Customer Name
- City
- Payment Method

These missing values were replaced with "**Unknown**" to maintain dataset consistency.

### 5.2 Duplicate Records

Duplicate rows were detected using Python and removed from the dataset.

### 5.3 Inconsistent Entries

The following inconsistencies were corrected:

| Before      | After       |
|-------------|-------------|
| karachi     | Karachi     |
| LAHORE      | Lahore      |
| islamabad   | Islamabad   |
| credit card | Credit Card |

## 5.4 Data Validation

The following data types were verified:

- Purchase Date → Date
- Quantity → Integer
- Unit Price → Numeric
- Discount → Numeric
- Total Amount → Numeric

Finally, the cleaned dataset was exported as **PakMart\_Retail\_Sales\_Cleaned.csv**.

## 6. Exploratory Data Analysis (EDA)

After cleaning the data, exploratory analysis was performed using Python.

The following analyses were completed:

- Monthly Sales Trend
- Top Selling Products
- Revenue by Product Category
- Revenue by City
- Customer Purchasing Patterns
- Payment Method Distribution
- Average Order Value
- Peak Sales Month

Different charts were generated using Matplotlib to better understand sales performance and customer behavior.

## 7. Dashboard Development

An interactive dashboard was created in Microsoft Power BI.

The dashboard includes the following KPIs:

- Total Revenue
- Total Orders

- Total Customers
- Average Order Value

The following visualizations were added:

- Monthly Sales Trend
- Top 10 Selling Products
- Revenue by Category
- Revenue by City
- Payment Method Distribution

Interactive slicers were also added for:

- City
- Product Category
- Payment Method
- Purchase Date

This allows users to filter and explore the data dynamically.

## **8. Key Insights**

The analysis revealed several important business findings:

- The dataset identified a peak sales month with the highest revenue.
- Smart Watch was the highest-selling product based on quantity sold.
- Electronics generated the highest overall revenue among all product categories.
- Sales performance varied across different cities, indicating stronger and weaker markets.
- Cash and Debit Card were among the most commonly used payment methods.
- Customer purchasing patterns showed higher demand for electronic products than other categories.

## **9. Business Recommendations**

Based on the analysis, the following recommendations are proposed:

1. Increase inventory for high-demand products such as Smart Watches and Smartphones.
2. Expand promotional campaigns in cities with higher sales performance.
3. Introduce discounts and marketing strategies for lower-performing product categories.
4. Encourage customers to use digital payment methods through cashback and promotional offers.
5. Use monthly sales trends to improve inventory planning and demand forecasting.

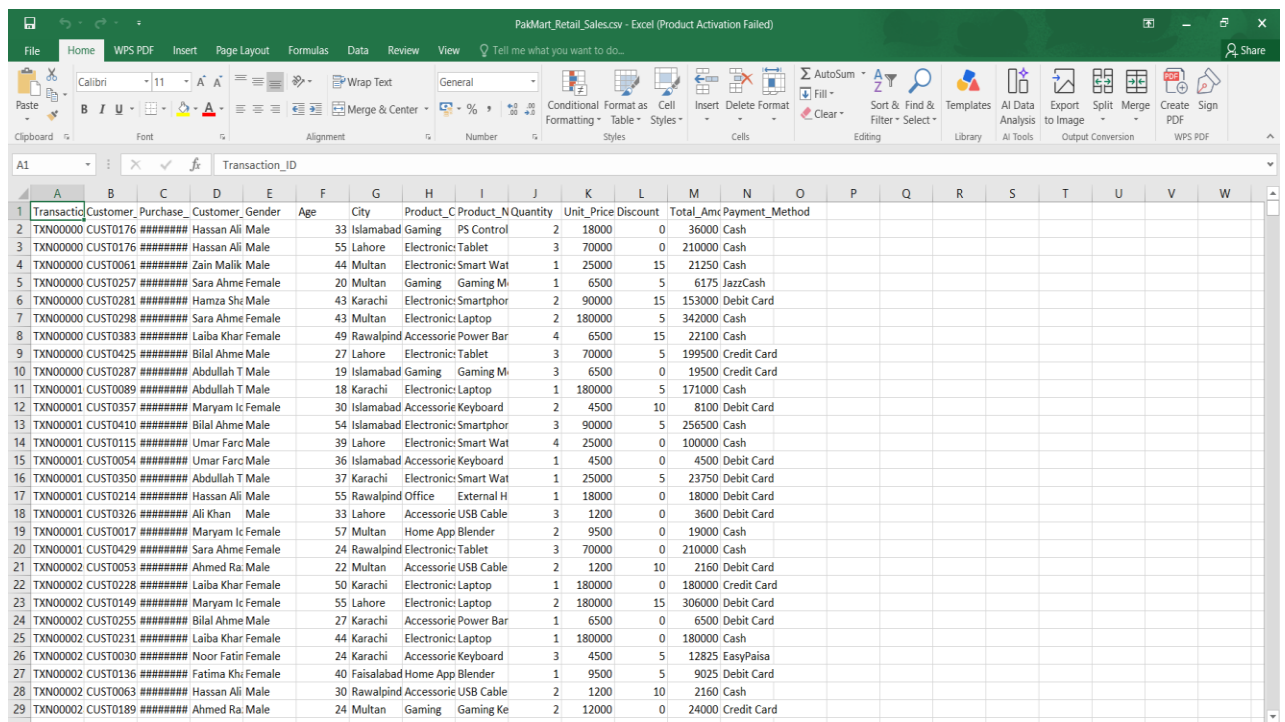
## 10. Conclusion

This project successfully demonstrated the complete data analytics lifecycle, including dataset creation, data cleaning, exploratory data analysis, business insight generation, and dashboard development.

The project provides a clear understanding of retail sales performance and customer behavior through interactive visualizations and business-focused analysis. The Power BI dashboard enables decision-makers to monitor KPIs, identify sales trends, and make informed business decisions. This project also strengthened practical skills in Python, Google Colab, data preprocessing, visualization, and business analytics.

## 11. Screenshots

- Raw Dataset Preview



| Transaction_ID | Customer | Purchase | Customer_Gender   | Age | City       | Product_C  | Product_N  | Quantity | Unit_Price | Discount | Total_Amc | Payment_Method |
|----------------|----------|----------|-------------------|-----|------------|------------|------------|----------|------------|----------|-----------|----------------|
| TXN00000       | CUST0176 | #####    | Hassan Ali Male   | 33  | Islamabad  | Gaming     | PS Control | 2        | 18000      | 0        | 36000     | Cash           |
| TXN00000       | CUST0176 | #####    | Hassan Ali Male   | 55  | Lahore     | Electronic | Tablet     | 3        | 70000      | 0        | 210000    | Cash           |
| TXN00000       | CUST0061 | #####    | Zain Malik Male   | 44  | Multan     | Electronic | Smart Wat  | 1        | 25000      | 15       | 21250     | Cash           |
| TXN00000       | CUST0257 | #####    | Sara Ahme Female  | 20  | Multan     | Gaming     | Gaming Mi  | 1        | 6500       | 5        | 6175      | JazzCash       |
| TXN00000       | CUST0281 | #####    | Hamza Shi Male    | 43  | Karachi    | Electronic | Smartphor  | 2        | 90000      | 15       | 153000    | Debit Card     |
| TXN00000       | CUST0298 | #####    | Sara Ahme Female  | 43  | Multan     | Electronic | Laptop     | 2        | 180000     | 5        | 342000    | Cash           |
| TXN00000       | CUST0383 | #####    | Laiba Khar Female | 49  | Rawalpind  | Accessorie | Power Bar  | 4        | 6500       | 15       | 22100     | Cash           |
| TXN00000       | CUST0425 | #####    | Bilal Ahme Male   | 27  | Lahore     | Electronic | Tablet     | 3        | 70000      | 5        | 199500    | Credit Card    |
| TXN00000       | CUST0287 | #####    | Abdullah T Male   | 19  | Islamabad  | Gaming     | Gaming Mi  | 3        | 6500       | 0        | 19500     | Credit Card    |
| TXN00001       | CUST0089 | #####    | Abdullah T Male   | 18  | Karachi    | Electronic | Laptop     | 1        | 180000     | 5        | 171000    | Cash           |
| TXN00001       | CUST0357 | #####    | Maryam Ic Female  | 30  | Islamabad  | Accessorie | Keyboard   | 2        | 4500       | 10       | 8100      | Debit Card     |
| TXN00001       | CUST0410 | #####    | Bilal Ahme Male   | 54  | Islamabad  | Electronic | Smartphor  | 3        | 90000      | 5        | 256500    | Cash           |
| TXN00001       | CUST0115 | #####    | Umar Farq Male    | 39  | Lahore     | Electronic | Smart Wat  | 4        | 25000      | 0        | 100000    | Cash           |
| TXN00001       | CUST0054 | #####    | Umar Farq Male    | 36  | Islamabad  | Accessorie | Keyboard   | 1        | 4500       | 0        | 4500      | Debit Card     |
| TXN00001       | CUST0350 | #####    | Abdullah T Male   | 37  | Karachi    | Electronic | Smart Wat  | 1        | 25000      | 5        | 23750     | Debit Card     |
| TXN00001       | CUST0214 | #####    | Hassan Ali Male   | 55  | Rawalpind  | Office     | External H | 1        | 18000      | 0        | 18000     | Debit Card     |
| TXN00001       | CUST0326 | #####    | Ali Khan Male     | 33  | Lahore     | Accessorie | USB Cable  | 3        | 1200       | 0        | 3600      | Debit Card     |
| TXN00001       | CUST0017 | #####    | Maryam Ic Female  | 57  | Multan     | Home App   | Blender    | 2        | 9500       | 0        | 19000     | Cash           |
| TXN00001       | CUST0429 | #####    | Sara Ahme Female  | 24  | Rawalpind  | Electronic | Tablet     | 3        | 70000      | 0        | 210000    | Cash           |
| TXN00002       | CUST0053 | #####    | Ahmed Ra Male     | 22  | Multan     | Accessorie | USB Cable  | 2        | 1200       | 10       | 2160      | Debit Card     |
| TXN00002       | CUST0228 | #####    | Laiba Khar Female | 50  | Karachi    | Electronic | Laptop     | 1        | 180000     | 0        | 180000    | Credit Card    |
| TXN00002       | CUST0149 | #####    | Maryam Ic Female  | 55  | Lahore     | Electronic | Laptop     | 2        | 180000     | 15       | 306000    | Debit Card     |
| TXN00002       | CUST0255 | #####    | Bilal Ahme Male   | 27  | Karachi    | Accessorie | Power Bar  | 1        | 6500       | 0        | 6500      | Debit Card     |
| TXN00002       | CUST0231 | #####    | Laiba Khar Female | 44  | Karachi    | Electronic | Laptop     | 1        | 180000     | 0        | 180000    | Cash           |
| TXN00002       | CUST0030 | #####    | Noor Fatin Female | 24  | Karachi    | Accessorie | Keyboard   | 3        | 4500       | 5        | 12825     | EasyPaiza      |
| TXN00002       | CUST0136 | #####    | Fatima Khi Female | 40  | Faisalabad | Home App   | Blender    | 1        | 9500       | 5        | 9025      | Debit Card     |
| TXN00002       | CUST0063 | #####    | Hassan Ali Male   | 30  | Rawalpind  | Accessorie | USB Cable  | 2        | 1200       | 10       | 2160      | Cash           |
| TXN00002       | CUST0189 | #####    | Ahmed Ra Male     | 24  | Multan     | Gaming     | Gaming Ke  | 2        | 12000      | 0        | 24000     | Credit Card    |

- Missing Values, Duplicate Records Check

```
[8] df.isnull().sum()
0s
Transaction_ID 0
Customer_ID 0
Purchase_Date 0
Customer_Name 11
Gender 0
Age 0
City 10
Product_Category 0
Product_Name 0
Quantity 0
Unit_Price 0
Discount 0
Total_Amount 0
Payment_Method 10
dtype: int64

[9] print("Duplicate Rows:", df.duplicated().sum())
0s
Duplicate Rows: 15
```

- Data Cleaning Output

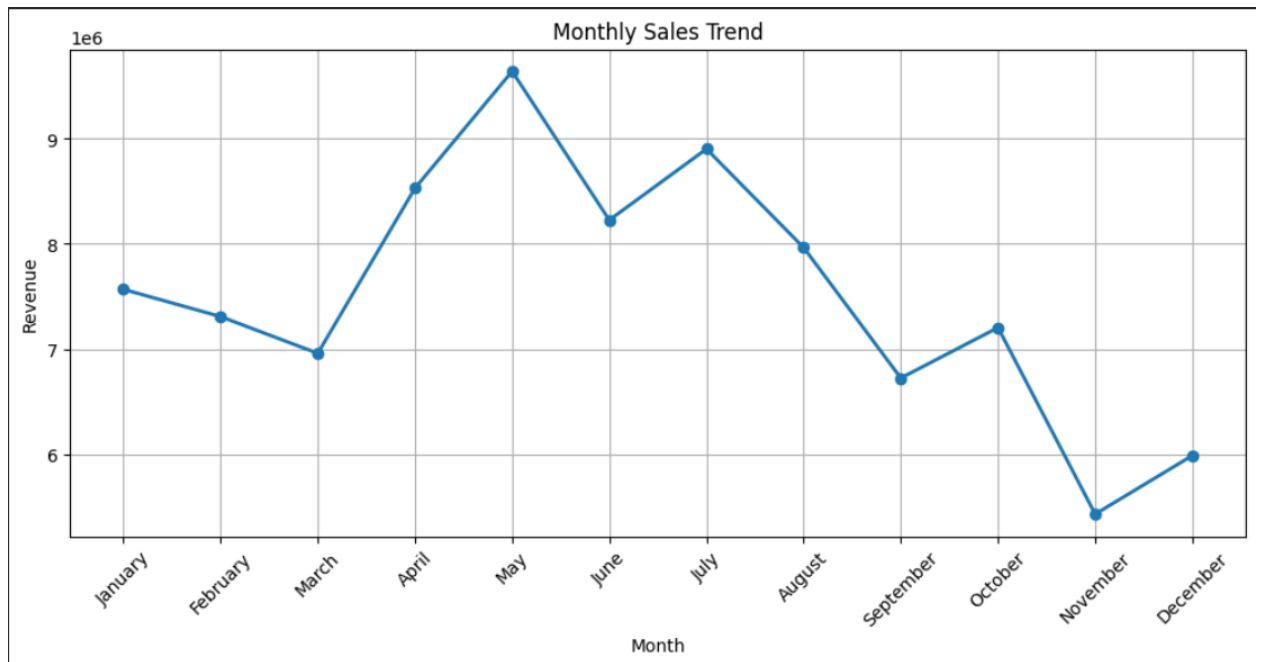
```
Transaction_ID 0
Customer_ID 0
Purchase_Date 0
Customer_Name 0
Gender 0
Age 0
City 0
Product_Category 0
Product_Name 0
Quantity 0
Unit_Price 0
Discount 0
Total_Amount 0
Payment_Method 0
dtype: int64

Removing Duplicate Rows

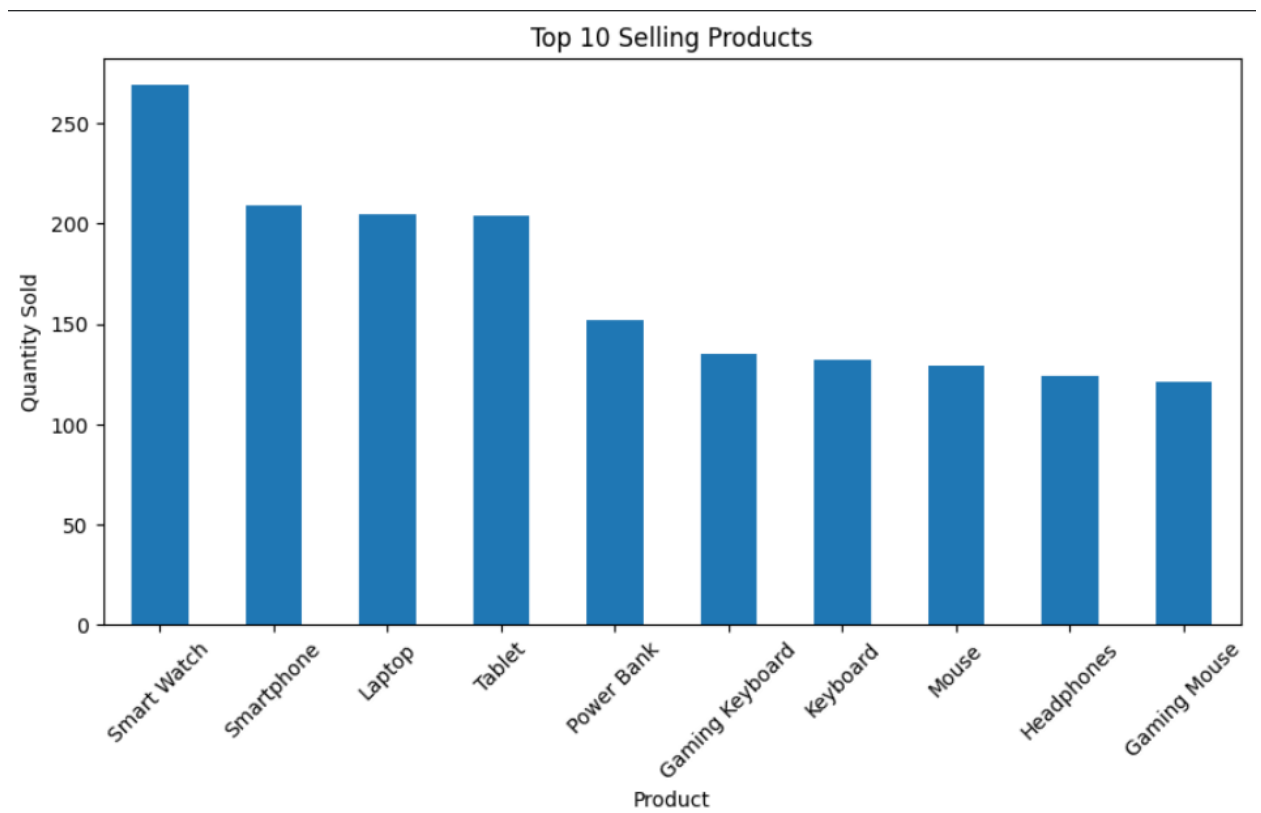
df = df.drop_duplicates()

print("Duplicate Rows:", df.duplicated().sum())
Duplicate Rows: 0
```

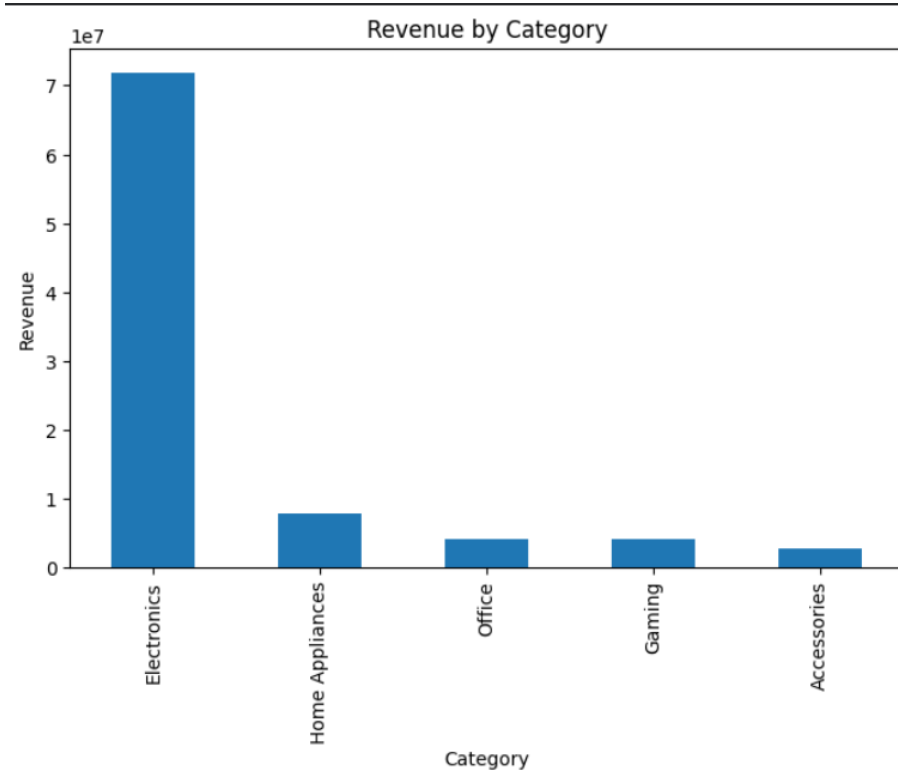
- Monthly Sales Trend



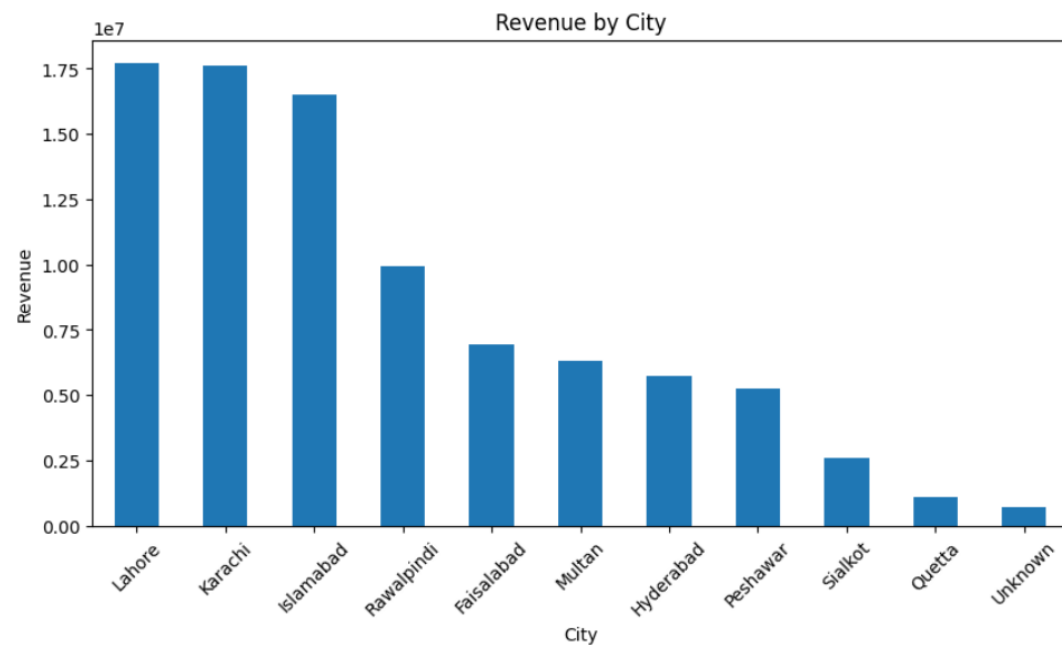
- Top Selling Products



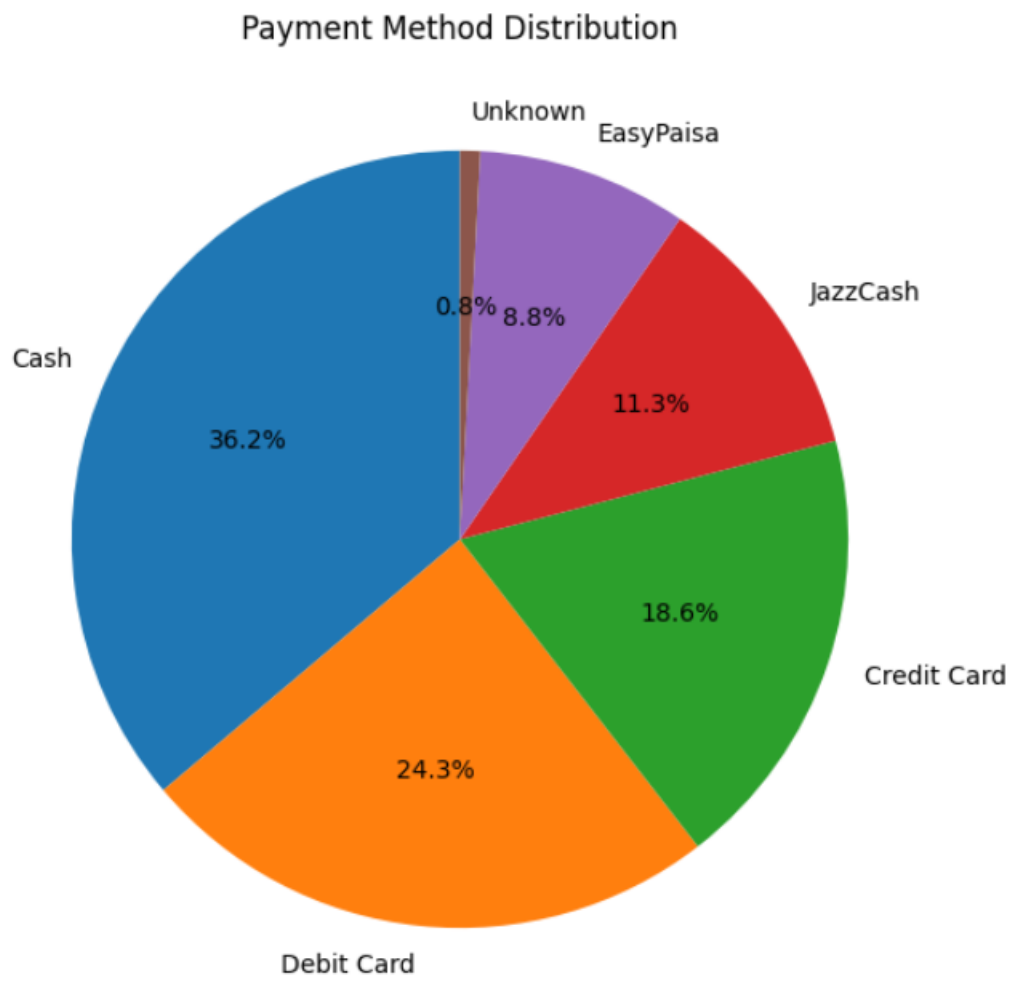
- Revenue by Category



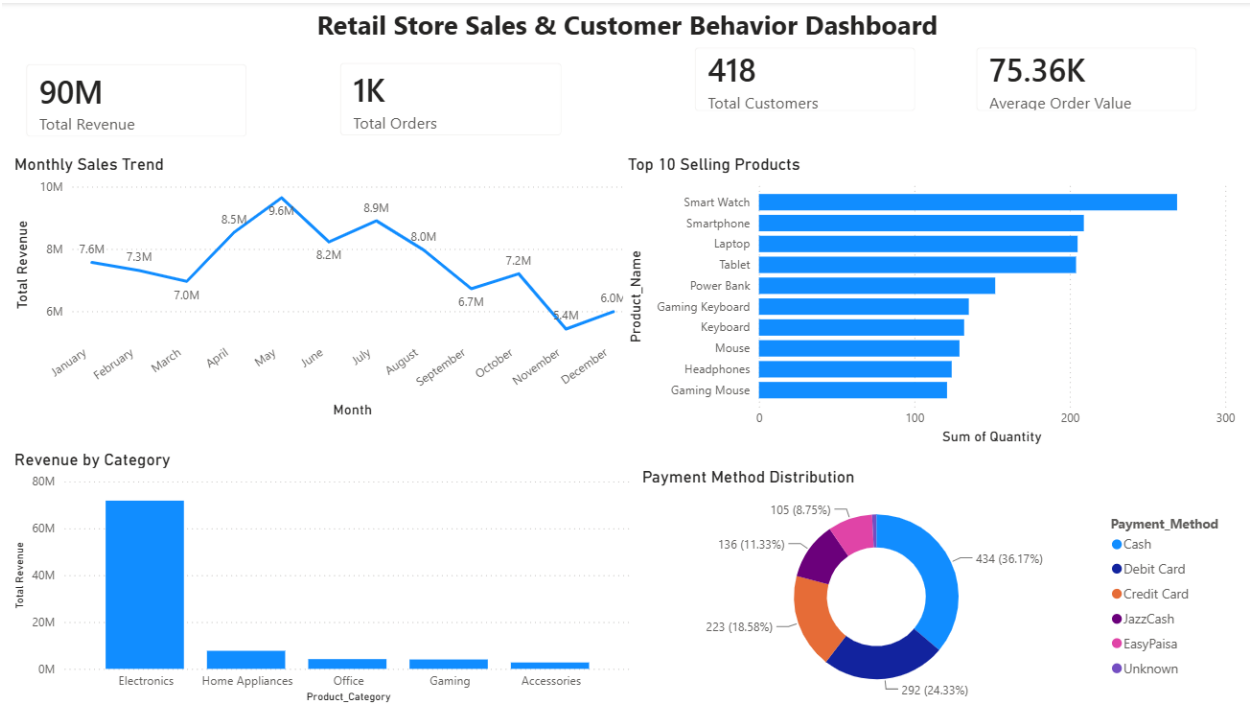
- Revenue by City



- Payment Method Distribution



- Power BI Dashboard (Page 1)



- Power BI Dashboard (Page 2)

